

method of lines for solving spatial models numerically

Lets simplify the advection-diffusion-reaction to a 1D problem
and assume diffusion is the only transport phenomena

$$\frac{\partial C_A}{\partial t} = -\nabla \cdot (vC_A) + D_{AB}\nabla^2 C_A + R_A.$$

↓ .. transform to only 1D (x)

$$\frac{\partial C_A}{\partial t} = -v \frac{\partial C_A}{\partial x} + D \frac{\partial^2 C_a}{\partial x^2} + R_A$$

↓ .. remove advection and reaction

$$\frac{\partial C_A}{\partial t} = D \frac{\partial^2 C_a}{\partial x^2}$$

C_a is the dependent state variable
 t is time variable (independent),
 x is spatial variable (independent)

A pipe is open in one end where the oxygen concentration is equal to the oxygen concentration in the air. The oxygen diffuses to the other end of the pipe where oxygen is instantaneously consumed, such that the concentration is always 0 at the right end



Define initial and boundary values.

$$\frac{\partial C_A}{\partial t} = D \frac{\partial^2 C_a}{\partial x^2}$$

Needs 1 initial value

Needs 2 boundary values

1) Replace only the spatial derivatives with finite difference approximations! (discretize the spatial derivative)

Finite differences from computer algorithms!

$$\frac{\partial C_A}{\partial t} = D \frac{\partial^2 C_a}{\partial x^2} \longrightarrow \frac{\partial C_A}{\partial t} = D \frac{C_{a,i+1} - 2C_{a,i} + C_{a,i-1}}{\Delta x^2}$$

i = index specifying position on x-grid,

Δx = spacing between grid points (assumed constant)

Central difference for 2. order

$$\frac{d^2 C_{A,i}}{dx^2} = \frac{C_{a,i+1} - 2C_{a,i} + C_{a,i-1}}{\Delta x^2}$$

Central difference 1. order

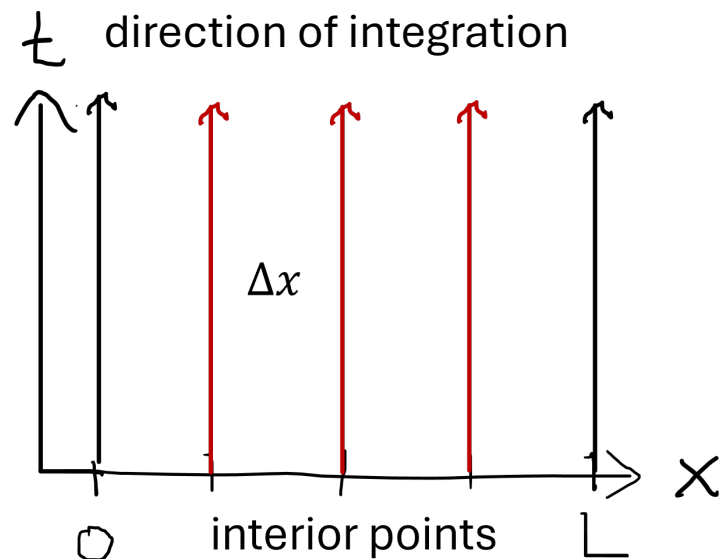
$$\frac{dC_{A,i}}{dx} = \frac{C_{A,i+1} - C_{A,i-1}}{2\Delta x}$$

Backward and forward difference 1. order

$$\frac{dC_{A,i}}{dx} = \frac{C_{A,i} - C_{A,i-1}}{\Delta x}, \quad \frac{dC_{A,i}}{dx} = \frac{C_{A,i+1} - C_{A,i}}{\Delta x}$$

3) discretize n domains and convert to system of ODEs

Example with 4 domains, 5 nodes



interior points

$$\frac{\partial C_{A,0}}{\partial t} = 0$$

$$\frac{\partial C_{A,0.25}}{\partial t} = D \frac{C_{a,x=0.5} - 2C_{a,x=0.25} + C_{a,x=0}}{\Delta x^2}$$

$$\frac{\partial C_{A,0.50}}{\partial t} = D \frac{C_{a,x=0.75} - 2C_{a,x=0.50} + C_{a,x=0.25}}{\Delta x^2}$$

$$\frac{\partial C_{A,0.75}}{\partial t} = D \frac{C_{a,x=1} - 2C_{a,x=0.75} + C_{a,x=0.50}}{\Delta x^2}$$

$$\frac{\partial C_{A,1}}{\partial t} = 0$$